

MATH 4 MACHINE LEARNING

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문제 1.

1-1) $P_1 = P_2 = P_3 = \dots = P_6 = \frac{1}{6}$

예컨대 $H(X) = \sum_{i=1}^6 -P_i \log_2(P_i) = 6 \times \left(-\frac{1}{6} \log_2\left(\frac{1}{6}\right)\right) = \log_2 6$

2) $P_1 \sim P_5 = 0, P_6 = 1$ 이므로,

$H(X) = -0 \log_2(0) \times 5 - \log_2(1) = 0$

3) $P_1 = P_3 = P_5 = \frac{1}{9}, P_2 = P_4 = P_6 = \frac{2}{9}$

$H(X) = -\frac{1}{9} \log_2\left(\frac{1}{9}\right) \times 3 - \frac{2}{9} \log_2\left(\frac{2}{9}\right) \times 3 = \log_2(1.9) \approx \frac{2}{3}$

4) $P_1 = 0.5, P_2 = P_3 = \dots = P_6 = 0.1$

$H(X) = -\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{10} \log_2\left(\frac{1}{10}\right) \times 5 = 0.5 - 0.5 \log_2(0.1) \approx 2.161$

1-2

X의 확률 질량 함수가 $P(X=x)$.

$H(X) = \sum_{x \in X} -p(x) \log_2(p(x)) \geq 0$

$H(X) = \sum_{x \in X} p(x) \log_2\left(\frac{1}{p(x)}\right) = \log_2(|X|) + \sum_{x \in X} p(x) \log_2\left(\frac{1}{p(x)}\right) - \log_2(|X|)$

$= \log_2(|X|) + \sum_{x \in X} p(x) [\log_2\left(\frac{1}{p(x)}\right) - \log_2(|X|)] = \log_2(|X|) - \sum_{x \in X} p(x) \log_2\left(\frac{|X|}{1/p(x)}\right) = \log_2(|X|) - \sum_{x \in X} p(x) \log_2(|X| p(x))$

따라서

$\sum_{x \in X} p(x) \log_2(|X| p(x)) = E[\log_2(|X| p(X))] \stackrel{\text{Jensen}}{\leq} \log_2\left(E[|X| p(X)]\right) = \log_2\left(|X| \sum_{x \in X} p(x)\right) = \log_2(|X|)$

따라서 $H(X) \leq \log_2(|X|)$

1-3. 1) $H(X) = -[0.6 \log_2(0.6) + 0.4 \log_2(0.4)] \approx 0.9110$

$H(Y) = -[0.55 \log_2(0.55) + 0.45 \log_2(0.45)] \approx 0.9850$

$\rightarrow H(X, Y) = -[0.45 \log_2(0.45) + 0.15 \log_2(0.15) + 0.1 \log_2(0.1) + 0.3 \log_2(0.3)] \approx 1.7814$

2) $H(X|Y) = H(X, Y) - H(Y) = 0.7964$

3) $I(X; Y) = H(X) + H(Y) - H(X, Y) = H(Y) - H(X|Y) = 0.1744$

4) $I(X; Y)$ 가 0 보다 크므로, 독립과 제한은 성립.

$I(X; Y) = 0 \rightarrow$ 독립성

$I(X; Y) = H(X)$ 는, Y값을 통해 X 확률에 대해

2-1

$$\lambda_1=1, \lambda_2=2, \lambda_3=2$$

$$(A-\lambda_1 I) v = 0 \rightarrow v = t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$(A-\lambda_2 I) v = 0 \rightarrow v = t \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + s \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \quad P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

2-2

$$B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \rightarrow \lambda_1=2, \lambda_2=1, \lambda_3=0 \quad Z = \begin{pmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$v = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -1 \end{pmatrix} \quad v^T = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$B = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

2-3

$$3-1 \quad L^{(n)} = -y^{(n)} \log(\hat{y}^{(n)}) - (1-y^{(n)}) \log(1-\hat{y}^{(n)}) \quad \text{Is Convex?}$$

$$\frac{\partial L}{\partial \hat{y}} = \hat{y} \log(\hat{y}) + (1-\hat{y}) \log(1-\hat{y})$$

$$\frac{\partial L}{\partial \omega} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial \omega} = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})} \cdot \hat{y}(1-\hat{y})x = (\hat{y} - y)x \quad \nabla L \rightarrow \text{is convex?} \quad \frac{\partial}{\partial \omega} ((\hat{y} - y)x^T) = \frac{\partial \hat{y}}{\partial \omega} x^T - \frac{\partial y}{\partial \omega} x^T$$

$$= (\hat{y} - y)(x x^T) \geq 0 \rightarrow \text{Convex}$$

$$3-2 \quad f(x) = x^4 - 2x^3 - 3x^2 + 4 \rightarrow f'(x) = 4x^3 - 6x^2 - 6x + 4$$

$$x_0 = 1, \quad \sigma = 0.2 \quad \text{is } x_{i+1} = x_i - \sigma \nabla f(x_i)$$

$$1) \quad x_1 = 1 - (0.2)(-7) = 2.4$$

$$\rightarrow x_2 = 2.4 - (0.2)(-0.9228) = 0.9228 \quad \leftarrow \text{2nd update}$$

$$3-3 \quad 1) \quad L(x, y, \lambda) = f(x, y) + \lambda g(x, y) = x^2 + y^2 + \lambda(x + y - 2)$$

$$2) \quad 1. \quad \frac{\partial L}{\partial x} = 2x + \lambda = 0 \rightarrow \lambda = -2x \quad \dots (1)$$

$$2. \quad \frac{\partial L}{\partial y} = 2y + \lambda = 0 \rightarrow \lambda = -2y \quad \dots (2)$$

$$3. \quad \frac{\partial L}{\partial \lambda} = x + y - 2 = 0 \rightarrow x + y = 2 \quad \dots (3)$$

$$3) \quad (1), (2) \rightarrow x = y \xrightarrow{(3)} x, y = 1 \xrightarrow{(1)} \lambda = -2$$

$$4) \quad f(1, 1) = 1^2 + 1^2 = 2 \rightarrow \text{is the minimum}$$